

WRO Engineering Manual

Team: The Redacted

July 21, 2026

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1 Selection of Components

1.1 Sensing

The WRO Future Engineers challenge sets certain criteria on the needed hardware. The second challenge makes teams not just detect walls but the pillars and their colors, as well as the ability to parallel park.

This means that a comprehensive and accurate inference of the playing field is required. There are multiple paths one can go down to fulfill this, there are perhaps only two practical routes. One is using distance sensors along with a camera and the other is using a 2-dimensional LiDAR.

The trade-offs between the two are fairly significant. The LiDAR comes at a higher cost than the camera and provides absolute and accurate cloud point data that can be used to detect the pillars and construct a map of the playing field with high accuracy. While the camera system is cheaper in the hardware cost, it has a significant cost in terms of compute. The detection of the actual pillars in the second round is easy, disregarding how slowly it would have to run and train (about $1/2 - 1$ Hz for good reliability on a wide FOV camera running YOLO, taking about 3 hours to train on 300 images, and a variable time to compute, not real time), lacks in the accuracy of positioning the pillars in dynamic environments. The LiDAR also provides very accurate and fast ($8 - 20$ Hz) feedback for absolute positioning on the go, something the camera does not. The aspect of parallel parking also plays into the advantage of the LiDAR. The LiDAR can provide a lot more detail than a distance assisted vision system could provide. If one did not use a LiDAR, at least 4 mutually perpendicular distance sensor, Time of Flight or Ultrasonic, would have to be implemented on an I2C bus.

All this made the case for using the LiDAR stronger. On smaller scales, where the camera is quite a bit larger than the vehicle and so is the LiDAR, using a LiDAR makes sense due to the current lack of compute density for very small and light boards being an issue. But, for larger vehicles, where the cost of cameras scales linearly and the LiDAR scales exponentially, using multiple cameras and cheaper, larger compute makes sense. Hence, we decided to use a LiDAR for our vehicle.

However, this did come at a major trade-off and design cost. We had to keep the whole vehicle shorter than 80 mm to accommodate for a clear view of the laser at about 90 mm and have a clearance of 10 mm from the top of the 100 mm arena walls to account for deviations in the walls and potentially the playing field as well.

For the actual hardware, the **YD-LiDAR X4 Pro** was chosen due to its compatibility with ROS and its relatively low cost and high scanning speed along with its resolution. The X4 Pro provides the high range that is needed, extending from 8 centimeters to 10 meters. When measuring less than 8 centimeters, parallel parking for example, the fall further away will be utilized to interpolate. The USB interface of the LiDAR also was very handy along with the openly available library and drivers.



Figure 1: The YD-LiDAR X4 Pro.

Because the LiDAR only outputs the point cloud data, and we need to find the color of the pillars as well, a wide FOV camera is used in tandem. It is only used to determine the color of the pillar and not used for detection. The LiDAR detects the pillar and the image from the camera is then used to assign the color. This keeps the compute costs very low and allows for very fast inference. The camera used is the **Arducam IMX219 Wide Angle Camera**. It has a field of view of 175° . This allows us to have the maximum window for inference possible as we don't need the data resolution a camera based solution would.

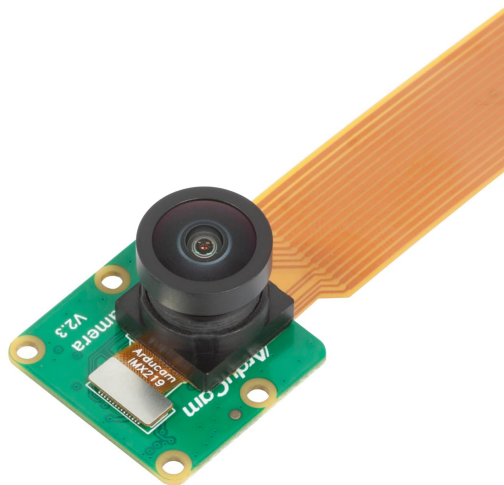


Figure 2: The IMX219 with a very wide angle lens.

1.2 Compute

The compute module doesn't need to be something excessive like the Jetson Orion Nano. The compute required to complete both the challenges is fairly low and could be done on an STM32 H7 micro-controller as well. The compute module has to fulfill the following requirements:

1. Small form factor.
2. Have a USB port.
3. Have a Camera Serial Interface .
4. Run ROS or Micro ROS.
5. Have a decent number of exposed IO pins for accessories.
6. Have the ability to read/write code for easy iterations.

This led to the Raspberry Pi 4B. Any other similar Single Board Computer (SBC) can be used. The 4GB RAM version was chosen so that a graphical user interface could be ran on it if needed when editing code during the competition. It would also provide sufficient headroom. A 2GB RAM version would have also been sufficient. A Raspberry Pi 5 would be overkill. The Pi 4 is also cheaper and easier to source. It would also have the compute needed to run the control loop at higher frequencies and have faster and more accurate control.

The only downside of the Raspberry Pi was its size. It's a lot larger than needed, and occupies crucial space in the limited volume. This was solved with intentional and clever design. Its availability of USB ports and the CSI makes it a very appealing solution. Custom hardware, ST based micro-controller or a board with the Raspberry Pi Compute Module, can be used but would be harder to source and would involve a longer development time. The Raspberry Pi also acts as a drop in solution as it has an external micro-SD card where all the code is stored, allowing for easy access and replaceability.



Figure 3: The Raspberry Pi 4 Model B.

1.3 Motion

A **rear wheel drive system** was chosen as it would distribute the complexity of the steering and drive systems, allowing easier manufacturing and assembly. The trade-off made here was that a **differential would be required** to prevent under steer that would be cause when turning at tight radii and slow speeds. This increases the complexity slightly, but not as much as using a drive and steering would need.

The motion system can be broken down into two parts, the drive system and the steering system.

1.3.1 Drive System

A brush less DC (BLDC) motor is proffered over a brushed motor due to it's higher resolution of speed changes particularly at a low speeds. That comes particularly handy when parallel parking. It is also preferred due to its efficiency and torque. The ideal case would be a stepper or BLDC motor with Field Oriented Control (FOC) as it would provide excellent position and torque control, but sourcing such FOC motor drivers in a small and compact size is very hard and expensive. A note should be made that developing such a custom FOC motor driver for 10-20A should not be hard but would take time.

For the motor, a Surpass Hobby **36 mm diameter in runner BLDC motor** is used. The 36 mm diameter is chosen as it provides enough **torque, acceleration and internal breaking** for a 1.5 kg vehicle. The length of the motor is 74 mm. This makes it a **3674 motor**. The motor is also widely used in scale RC cars and hence is easy to source. The motor is rated at **1700 kv**, which provides generous speed and high torque.



Figure 4: The Motor.

A Surpass Hobby 3 Phase **120A 4S ESC** is used. This Electronic Speed Controller is pre-

ferred due to its high **reliability and headroom, along with its configurability**. The ESC can be configured to operate at certain speed, torque and braking levels with a configurator device.



Figure 5: The ESC.

Do note that if built again, a shorter motor and lower ampere rated ESC would be used, something like a 3650 motor and a 60A ESC. The 3674 motor is too long and has excessive torque that is not really needed. The shorter motor should be used to reduce the weight as much as possible. The ideal motor would be a 3650 motor rated somewhere about 1000 kv. The 60A ESC would also be chosen to reduce weight. The current setup was chosen because it is what was available without purchasing new components.



Figure 6: The Configuration Device that connects via PWM.

1.3.2 Steering System

A **direct drive steering system** was chosen for the vehicle, meaning that the servo directly drives the wheels. This was chosen as the steering links would be manufactured using 3D print-

ing and each joint would cause additional backlash in the linkage mechanism. A system with a grater mechanical leverage would hence be more complex and less accurate, while taking more volume.

The **S2500M** servo motor is used, which provides a torque of about $25 \frac{kg}{cm}$ at about 8V. This would provide ample headroom in a direct drive steering system. The only downside to contend with in this system is its weight and the fact that the servo runs on 8V, meaning it will need its own power supply. Do note that a lower torque servo that operates on 5V can be used if needed, the S2500M was available without purchasing new components.



Figure 7: The S2500M metal gear servo motor.

1.4 Power Management

The batteries were chosen on strict power consumption metrics. [Power usage table].

To provide adequate headroom for the 3+ minutes of operation needed, **2 DOGCOM 4S 850 mAh batteries were used in parallel.** A single 4S 850 mAh battery in testing at full motion capabilities provided power for about 2.5 minutes before it went below the safe discharge voltage of 3.5 V per cell. Hence a second battery is needed for headroom.

The 4S batteries are used as they have the maximum voltage compatible with the setup, allowing the maximum acceleration and torque for the vehicle. The batteries use a **Li-Po (Lithium Polymer)** chemistry as that provides the high discharge current needed. It is also widely available and hence easy to source. On a larger scale vehicle, Li-Ion cells would be used due to their longer discharge cycle rating as the only major draw back with Li-Po batteries are their lower charge cycle rating.



Figure 8: The DOGCOM 4s 850 mAh battery.

1.5 The Additional Hardware

Additional hardware such as bearings and wheels are needed to build the vehicle. These are things that cannot be 3D printed and are manufactured in other ways or are sourced.

1.5.1 Bearings

For the two front steering knuckles, a **thrust roller bearing** is used. This is a type of bearing that combines a needle and thrust bearing in a single housing. It is used to constraint the location of the two front steering wheels though the axis of rotation and ensure there is no play present. If a normal roller ball bearing would be used, the outward and asymmetric torque of the wheel "pin" would misalign the groove and the balls causing excessive play. The thrust roller bearing is a compact solution used in real cars as well.

The inner diameter of the bearing is 12 mm. It is the "**BCWN-T-d12-IR**" bearing, sourced from JLCMC.

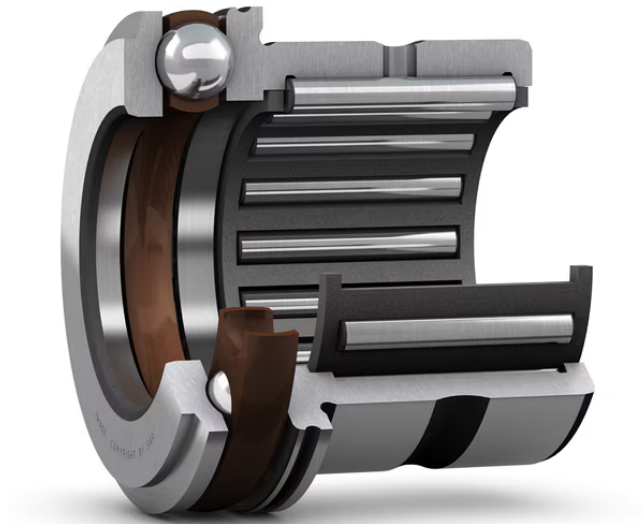


Figure 9: The internals of a Combined needle roller / thrust ball bearing.

For the differential, traditional roller ball bearings were used as the axial forces transferred are not that high. The **6901ZZ** ball bearing is used as it is in a very compact package that has an internal diameter of 12 mm, an external diameter of 24 mm and a width of 6 mm, making it ideal to put in the already compact differential gear box.



Figure 10: The 6901ZZ roller ball bearing.

To support the rear axle, needle roller bearings are used. They are used as they have low friction and can withstand very high forces perpendicular to the axis of rotation, perfect for transferring the weight from the main body of the vehicle to the wheels. The "**BCWG-NK12/12**" bearing was used, sourced from JLCMC. It has an inner diameter of 12 mm, an

outer diameter of 19.5 mm and is 12 mm deep.



Figure 11: The Needle Roller bearing.

1.5.2 Carbon Fiber Tubes

Carbon fiber tubes were used at several places as axles, linkage connectors and structural connectors. This decision was made as carbon fiber has a very high tensile strength and is particularly very stiff perpendicular to its length.

A combination of 2 diameters were used. The axle for the rear two wheels were made using **12 mm outer diameter** and 10 mm inner diameter carbon fiber tubes. Any aluminum or steel shaft could be used here, but a 3D printed shaft cannot be used as the layer lines of the print would be perpendicular to the layer lines, leading to very weak prints. Carbon fiber was used as it is light weight and relatively easy to source in **roll wrapped rods**. This tube consists of "twill" and not "twe".



Figure 12: The 12 mm CF rod.

The other rod used was a **3 mm outer diameter solid pultruded** rod. It is used in the steering links as well as the structural spin that runs throughout the vehicle to provide stiffness to the frame and not let the printed parts sag. It is pultruded, meaning it does not consist of carbon fibers but of chopped carbon bits held by epoxy. It is comparatively less stiff but still very strong. It consists of "toe" and no "twill".



Figure 13: The 3 mm PultrudedCF rod.

2 PCB Assembly

The first question one should ask is why use a PCB at all? Why not just use jumper wires and other modules?

The answer is quite simple yet complex. A Printed Circuit Board (PCB) integrates all the power management components needed into a sleek and compact form factor without a lot of wiring. The Raspberry pi has a 20×2 pin interface, which isn't very neat in it self. One can connect the individual jumpers to the header pins but it becomes VERY cumbersome very fast. Hence a PCB is almost necessary for a vehicle that can be easily serviced and is well engineered. Another major driver for this, as you will learn, comes from issues learnt from Siddesh's participation in WRO the previous year. KiCAD was used as the eCAD tool to design the PCBs.

2.1 Purpose

When participating in WRO 2025, the SD Card we were using failed on us the day before the final competition. It was corrupted and the Raspberry Pi would not boot up at all. Turns out the issue was that the Raspberry Pi was constantly undervolting under load. We had attempted to add a capacitor before the competition to resolve the issue, but the effort proved futile. Upon reviewing the logs, we confirmed that these undervoltage events were, in fact, what caused the SD card to fail. The buck converter was rated for 5V at a nominal current of 3A and a burst current of 5A. Yet the Raspberry Pi wouldn't boot.

The Raspberry Pi needs a voltage higher than just 5V. It needs a voltage of about 5.15V to prevent the undervolting issue we had seen earlier. We searched the market for a buck converter capable of providing the 5.1V and 3A but I couldn't find a suitable option. Hence it was determined that a custom solution was needed.

Hence the following considerations for the PCB were formed $\rightarrow$

1. Deliver 5.15V to the Raspberry Pi and have a continuous rating of atleast 3-5A to power the LiDAR as well.
2. Deliver 8V to the servo motor at 2A.
3. Provide a better hardware interface than the 20×2 connector for the Servo and ESC.
4. Be compact and reliable.
5. Have additional expansion ports for Time Of Flight Sensors, and more Servos.
6. Integrate the power switch and start switch, along with indicator LEDs needed for the rules on to the board.
7. Look good!

2.2 Form Factor

The first thing to decide after deciding the purpose is to decide the form factor of the PCB. This in turn, will decide where the PCB will live.

The best form factor is having the PCB as a hat above the Raspberry Pi itself. This is because of the following reasons →

1. There is a relatively large space above the Raspberry Pi, somewhere about $65\text{ mm} \times 56\text{ mm}$.
2. A simple header pin board can be added to the PCB, meaning it will directly interface with the header pins on the Raspberry Pi. This would eliminate the need for adding wires between the Pi and the PCB, ensuring efficient and smooth power distribution and connections.
3. It would provide ease of maintenance to the PCB and uninterrupted access to the switches.

2.3 Selection of the ICs

Due to the custom need for the 5.15V and a lack of existing modules, a custom voltage converter module would have to be implemented on the PCB hat. Since we were already using SMD (Surface Mount Device) components for the 5.15V PCB, we decided to integrate the 8V part as well on the PCB itself rather than have a separate module.

We use buck converters, which are a form of switching regulators rather than using linear regulators. Although linear regulators provide the smoothest possible voltage, they dissipate the remaining energy as heat, and hence are very inefficient and only good for low power applications where the voltage difference between the input and output is very low. However, we need the exact opposite. We need to draw an insignificant current and our voltage difference between the input and output is substantial. Hence we use switching regulators. The only downside is that they do not have a clean voltage output as they use a MOSFET to switch at between 0V and battery voltage at high frequencies. This can be compensated for using good design and with adding filtering capacitors at the output.

2.3.1 The 5.1V Rail

The LM61460 from Texas Instruments was chosen as the switching regulator due to its size and the fact that it has a continuous limit of 6A. It is also ideal because it has symmetric switching nodes and in built MOSFETs, which reduce noise.

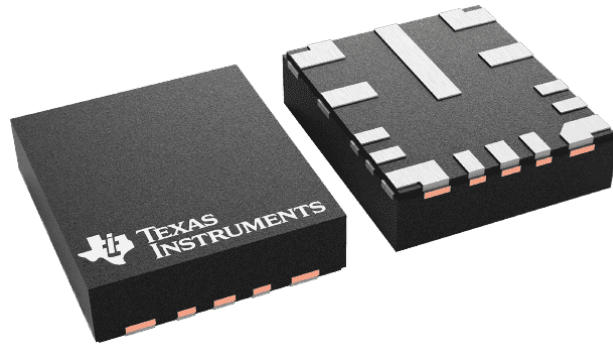


Figure 14: The LM61460 in the VQFN-HR package.

The values for the schematic were obtained using the TI Workbench.

2.3.2 The 8V Rail

The TPS62932 from Texas Instruments was chosen as it is incredible small and has a current rating of 2A. It also has an in built MOSFET in its incredible tiny package of just $2.1 \text{ mm} \times 1.6 \text{ mm}$.

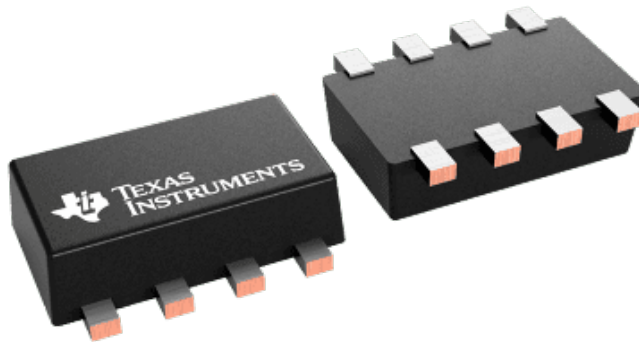


Figure 15: The TPS62932 in the SOT-583 package.

The values for the schematic were obtained using the TI Workbench.

2.4 The Schematic

The first stage of the PCB development process is creating the schematic; the wiring diagram for all the connections. The first step is to add all the individual components that are used, the Raspberry Pi, the buck converter ICs, the motors, the LEDs, the switches, the power connectors and any other I/O that maybe needed.

The switching regulators need additional hardware, such as resistors and capacitors along with an inductor. The value of these components decide what voltage you get at the output. To find these values, the TI workbench tool was used. The TI workbench allows one to configure the parameters and then produces a schematic (Figure 16). This is then used to set the value and the connections of the SMD components.

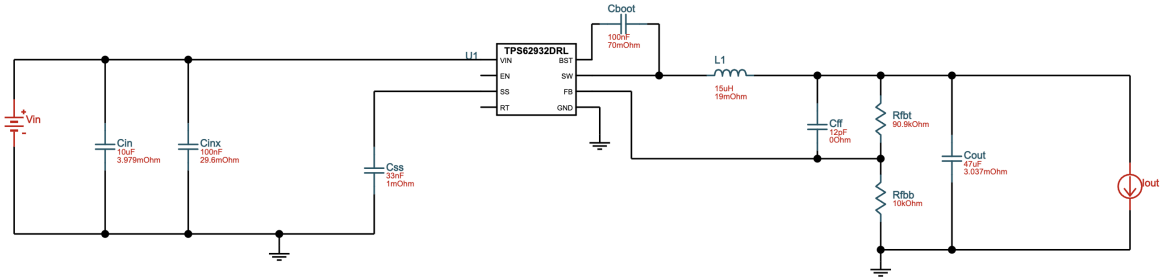
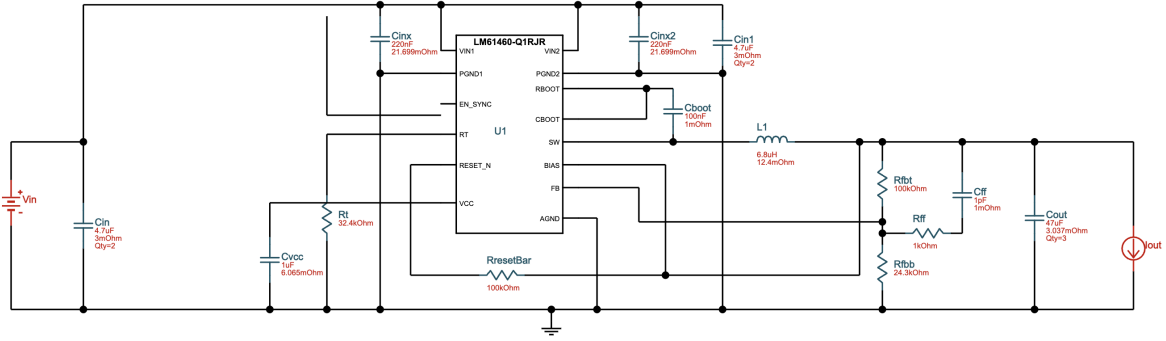


Figure 16: The files for the schematic given by TI Workbench.

The schematic (Figure 17) created is in 3 pages. The main page has the XT30 power connector input and the power switch. The Raspberry Pi page consists of the Raspberry Pi, the sensors and I/O devices. The power page consists of all the power regulation circuitry. It also has 2 additional electrolytic capacitors, one $220\mu\text{F}$ and the other $270\mu\text{F}$. They both act as decoupling capacitors for the Raspberry Pi and the servo respectively. They ensure that when there are load spikes from either devices, the Raspberry Pi booting or the servo starting

to move, the spike does not reach the ICs and is handled locally.

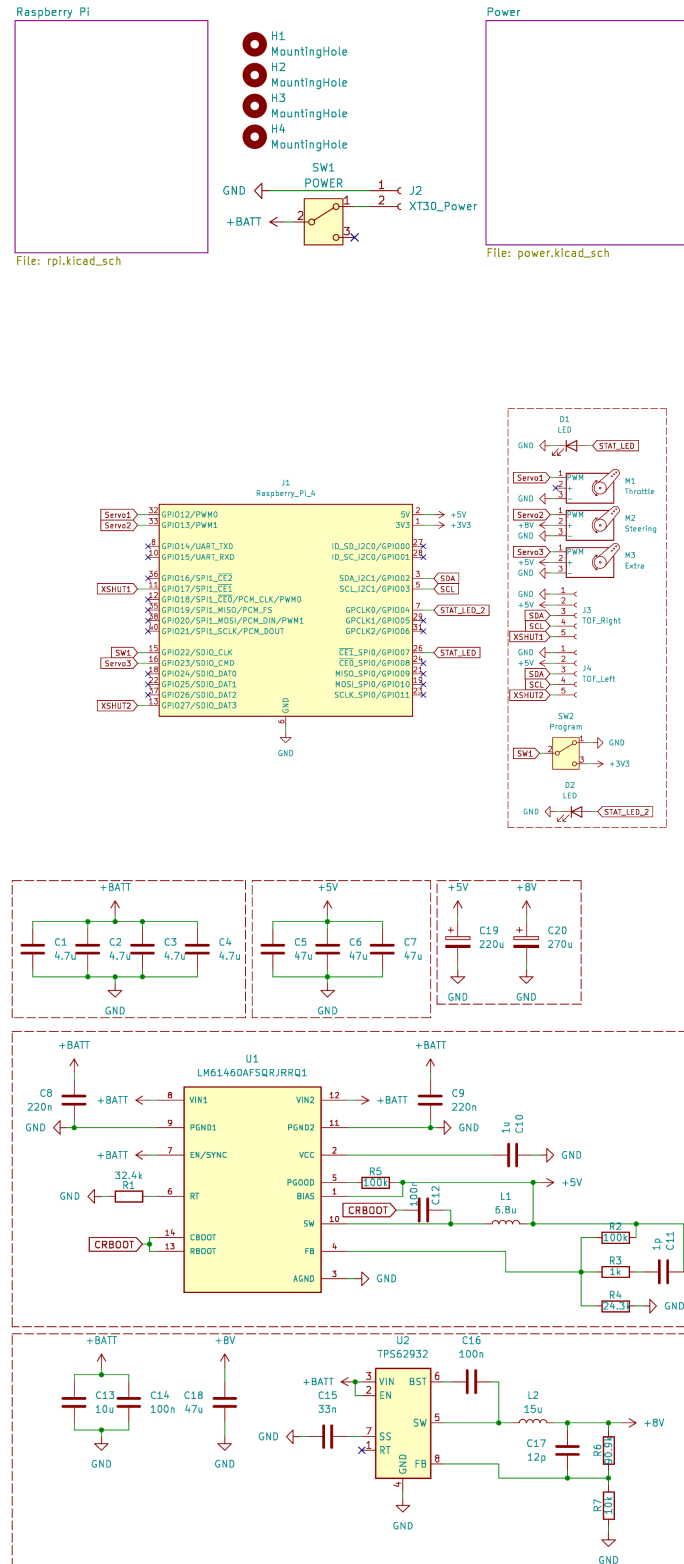


Figure 17: The 3 pages of the schematic.

The PDF for the Schematic can also be found in the PCBA folder. The BOM for all the parts is also available there, along with the manufacturer used.

2.5 The Layout & Routing

This step is very crucial, it is responsible for how well the components interact with each-other and determines how clean your output voltage is. A **4 layer board** was chosen. This was done as it would make the routing easier across multiple layers and because a 4 layer stack up is recommended in the LM61460 data sheet.

All the SMD components would be place on one side of the board so that the board could be assembled on a hot plate. This would also aid with the ease of assembly. The following is the stack up →

1. Layer 1: It carries much of the important **signals**, especially for the buck converters.
2. Layer 2: A solid **Ground** plane to reduce EMI and act as a fast return path of the top player.
3. Layer 3: The **Power** plane consisting of the input voltage, 5.15V and 8V.
4. Layer 4: A **Ground** plane that also carried some of the **signals** from the Raspberry Pi.

The first thing to do is create the board outline. A rounded rectangle with length 65mm, height 56mm and a corner radius of 2mm was chosen. A 6mm notch was then removed from the bottom right of the board to allow the camera ribbon cable to pass through. Figure 18 shows the shape of the board.

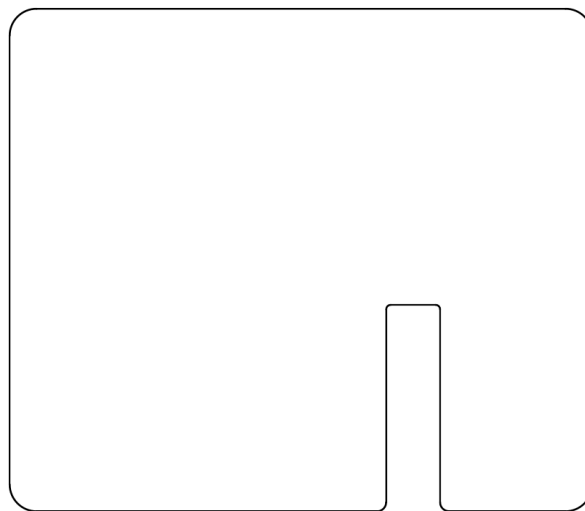


Figure 18: The "edge cuts" layer.

After this, the fixed components are place, such as the mounting holes for the board and the 20×2 female header pin that connects to the Raspberry Pi. These need to be incredibly precise as they are the locating hardware for the hat to connect on to the Raspberry Pi. M2.5 holes are used.

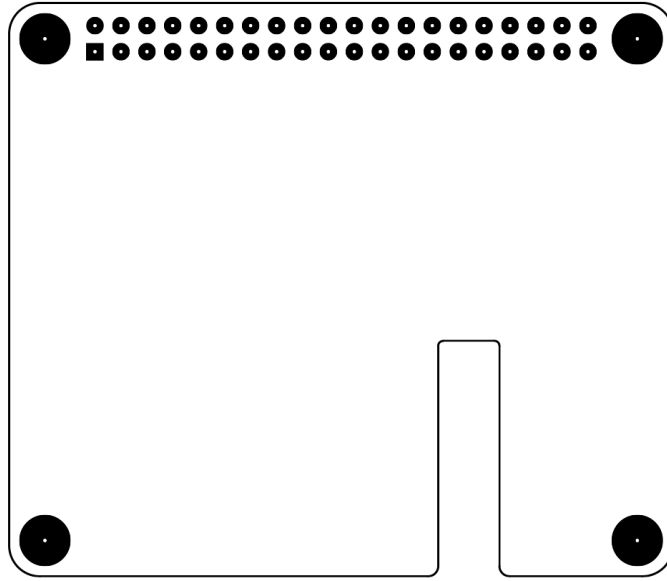


Figure 19: The fixed items layer.

The next part is the most cumbersome. All the physical parts need to be laid out in a manner that ensures the most ideal connections possible. The 3 main current loops; ON-State Current Loop, OFF-State Current Loop and The "Hot Loop"; of the buck converter need to be kept as short as possible. Since we are using a symmetric buck converter, it also needs to be made sure that the power lines to both sides are as similar as they can be. The path between the switching node and the inductor needs to be as short as possible. All the connectors need to be accessible.

For this reason, the power connector (XT30) and the power switch were placed on the right side. The switching circuitry was spread around the middle. The second switch and the connectors for the motor and servo were kept on the left side. The electrolytic capacitors were placed as close to the outputs as possible. The LEDs were also placed in the middle. The extra components, such as the connectors for the 2 ToF sensors and an extra PWM connector were kept on the bottom and bottom right respectively. Figure 20 shows the layout of the final board. It is fairly cramped but a lot more items, such as more LEDs, Current and Voltage sensing and a CAN BUS port can also be added in the future.

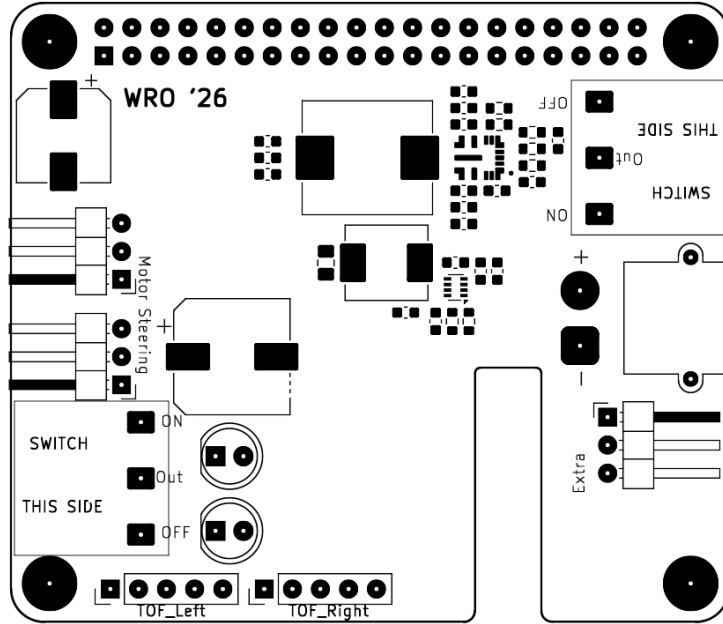
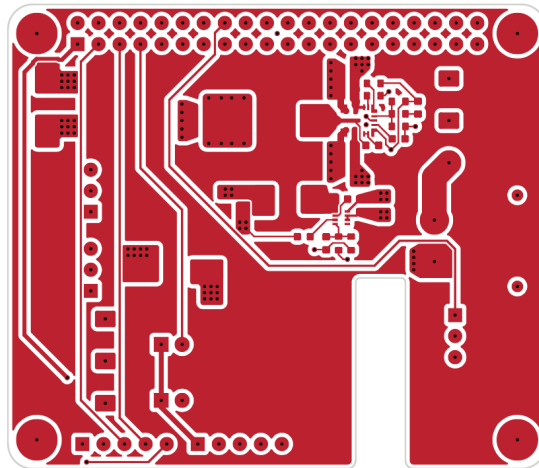


Figure 20: The final layout depicting the placement of all the components.

Figure 21 shows the actual copper layers as they are. Copper pours are preferred whenever possible. They also help with thermal management. It was made sure that there were no wires beneath the two inductors to make sure all the signal traces are as clean as possible.



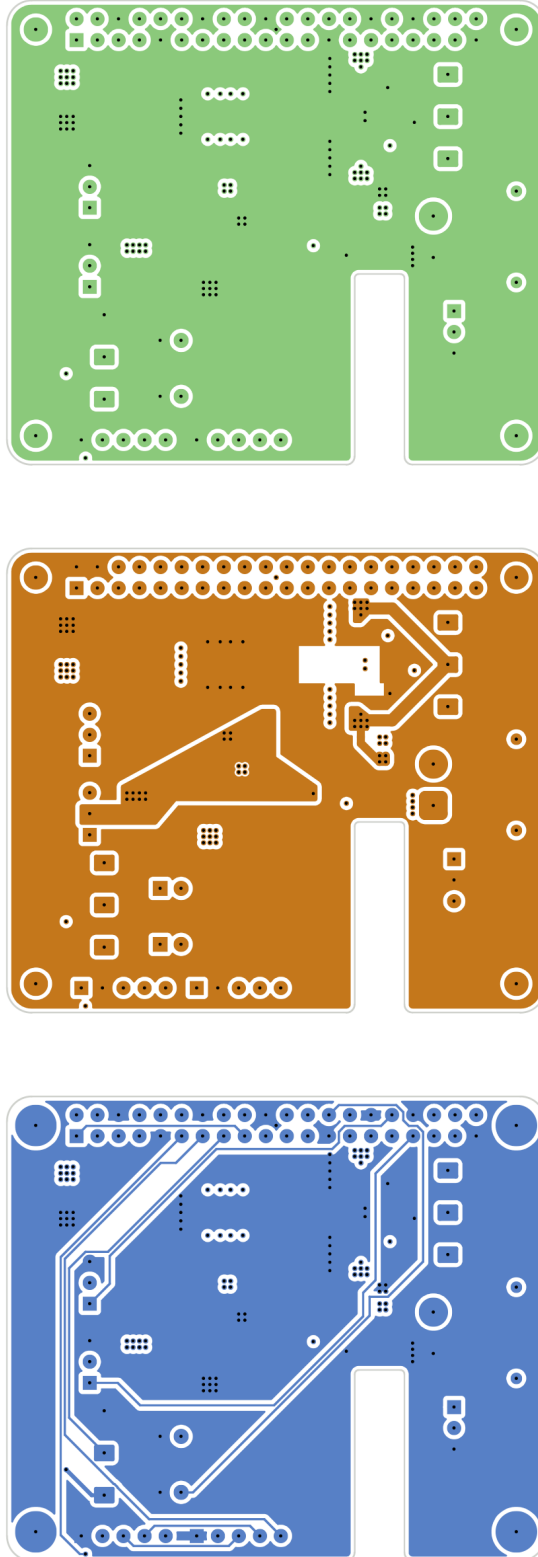


Figure 21: The 4 copper layers. Layer1 $\rightarrow$ 4.

Some art work and quotes were added to the free space to ensure complete utilization and occupation of left over space. A black solder mask was chosen. Figure 22 show the PCB rendered in the KiCAD 3D viewer. The contribution for the art work goes to M.C. Escher.

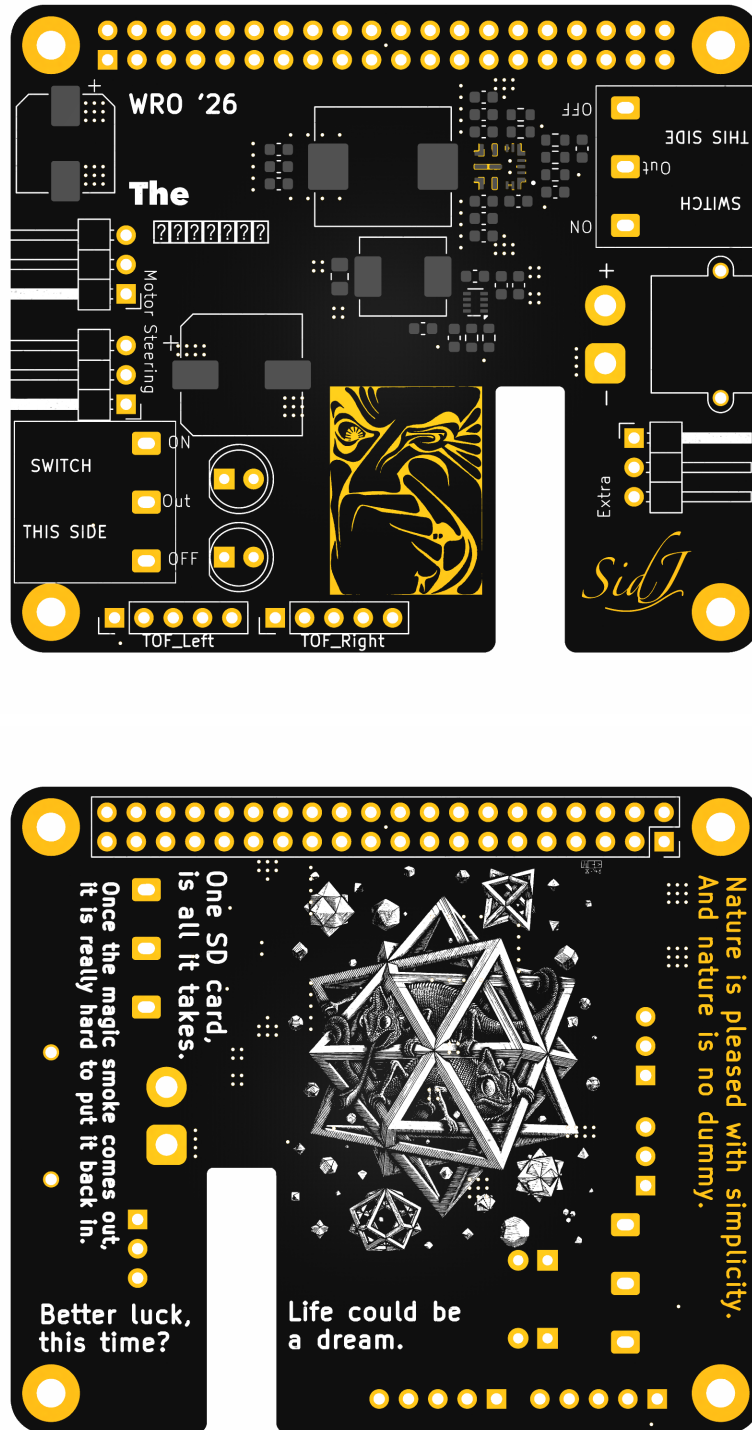


Figure 22: The rendered PCB.

The PCB was manufactured by JLC. JLC was chosen due to their lower cost for black solder mask compared to Indian manufactures. The following are the basic parameters:

1. Base material: FR4
2. PCB Thickness: 1.6mm

3. PCB Color: Black
4. Silkscreen: White
5. Material Type: FR4 TG135
6. Surface Finish: Lead Free HASL (Hot air solder leveling)

2.6 Assembly

The assembly was done by hand. 0603 class components are not too hard to place by hand. Alternatively, assembly can be done by the manufacturer as well.

A stencil was used to apply the layer of 183° C solder paste on to the board with the help of an old credit card. The "Interactive Html Bom" plugin (Figure 23) was used to aid with assembly. The plugin creates a interactive html file that can be used to figure out the placement of the components. Alternatively, the PCB and schematic editor can itself be used to figure the placement out but using the plugin is recommended do to its convenience. Once the components have been laid out, a final visual inspection can be carried out.

The boards can then be re-flowed on the hot plate. It is recommended to follow the heat curve of the solder paste being used. If the plate available does not have the ability to follow a curve, let the board heat soak at about 50-80° C below the melting point of the solder paste for about 1-2 minutes to let the flux activate before increasing the temperature to its final point.

Our hot plate did not have the feature to follow the heat curve so we let our boards soak at 130° C for 90 seconds after increasing the temperature to 195° C till all of the paste re-flowed and waited 30 seconds more before taking the boards off the hot plate.

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Ref lookup Filter

Source	Placed	References	Value	Footprint	Quantity
1	☐	C1, C2, C3, C4	4.7u	C_0603_1608Metric	4
2	☐	C5, C6, C7	47u	C_0603_1608Metric	3
3	☐	C12, C14, C16	100n	C_0603_1608Metric	3
4	☐	C8, C9	220n	C_0603_1608Metric	2
5	☐	C10	1u	C_0603_1608Metric	1
6	☐	C11	1p	C_0603_1608Metric	1
7	☐	C13	10u	C_0603_1608Metric	1
8	☐	C15	33n	C_0603_1608Metric	1
9	☐	C17	12p	C_0603_1608Metric	1
10	☐	C18	47u	C_0805_2012Metric_Pad1.18x1.45mm_HandSolder	1
11	☐	C19	220u	CP_Elec.8x10	1
12	☐	C20	270u	CP_Elec.10x10	1
13	☐	R2, R5	100k	R_0603_1608Metric	2
14	☐	R1	32.4k	R_0603_1608Metric	1
15	☐	R3	1k	R_0603_1608Metric	1
16	☐	R4	24.3k	R_0603_1608Metric	1
17	☐	R6	99.9k	R_0603_1608Metric	1
18	☐	R7	10k	R_0603_1608Metric	1
19	☐	L1	6.8u	L_Chilisin_040708101040	1
20	☐	L2	15u	L_Chilisin_040808060618	1
21	☐	D1, D2	LED	LED_05.8mm	2
22	☐	U1	LM5146BASF5QJ3RQ1	IC_LM5146BASF5QJ3RQ1	1
23	☐	U2	TPS62932	SOT-583-8	1
24	☐	SW1	POWER	Switch	1
25	☐	SW2	Program	Switch	1
26	☐	M1	Throttle	PinHeader_1x03_P2.54mm_Horizontal	1
27	☐	M2	Steering	PinHeader_1x03_P2.54mm_Horizontal	1
28	☐	M3	Extra	PinHeader_1x03_P2.54mm_Horizontal	1
29	☐	REF**	perfume	perfume	1
30	☐	J2	XT30_Power	AWASS_XT30PW-M_1x02_P2.50mm_Horizontal	1
31	☐	J3	TOF_Right	PinHeader_1x05_P2.54mm_Vertical	1
32	☐	J4	TOF_Left	PinHeader_1x05_P2.54mm_Vertical	1

Figure 23: The Interactive HTML BOM Tool.

The HTML file generated by the plugin can be found in the PCB folder along with the BOM.

2.7 Mistakes & Errors

3 3D Prints & CAD